

MVC

Executive Summary

MVC Composite Dielectric Material

Meta-Void Composite Laminate — Architecture Upgrade V5.0

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Sources: TECHNICAL_SPECIFICATION.md

CARRIER-TO-BOARD_INTERCONNECT_DOSSIER_EN.pdf

MASTER_DOSSIER_ARCHITECTURE_UPGRADE_V5.0_EN.pdf

2.98

Dk @ 10 GHz

0.0034

Df @ 10 GHz

-71%

Cost vs Rogers

AGPL+CERN

Open-Source

Intended audience: Engineering managers, procurement leads, and executive stakeholders

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Purpose & Background

The MVC (Meta-Void Composite) programme delivers an open-source, drop-in dielectric laminate designed to replace Rogers RO4003C-class and PTFE-based substrates in RF, mmWave, and high-speed digital applications. Version 5.0 represents a production-ready release, co-authored with interconnect architecture data from the Carrier-to-Board Interconnect Dossier and the Master Architecture Upgrade V5.0 — both canonical references in this package. All design files, specifications, and processing data are released as open-source hardware and software under AGPL-3.0 and CERN-OHL-S, enabling broad industry adoption without licensing fees or supply-chain lock-in.

Key Benefits at a Glance

Low Dielectric Loss • Dk = 2.98 @ 10 GHz • Df = 0.0034 @ 10 GHz • 16% lower Dk than RO4003C • 20% lower loss vs Isola I-Speed	Dimensional Stability • CTE X-Y = 14 ppm/C (matched to FR4) • Tg = 185 C — wide process window • HAD < 5 µm — precise via registration • IST <10% dR @ 500 thermal cycles
Manufacturing Compatible • Standard PCB lamination (175–185 C) • UV 355 nm and CO2 laser via drilling • ENIG / HASL / Immersion Ag qualified • Drop-in for existing PCB tooling	Open-Source & Cost-Effective • 1.0x cost index vs 3.5x for RO4003C • AGPL-3.0 + CERN-OHL-S licensed • No royalties, no supply lock-in • Full specs, test data, and SEM data included

Cost Impact & Silicon-Reduction Savings

Adopting MVC in place of Rogers RO4003C delivers material cost savings of approximately **71 %** per panel at equivalent panel size, based on 2025 bulk pricing. The lower dielectric loss (Df 0.0034 vs 0.0027 for RO4003C at 10 GHz) remains within the acceptable performance envelope for most mmWave front-end modules while providing substantial bill-of-materials relief. Critically, the Dk reduction (2.98 vs 3.55) allows trace widths to be narrowed by **~8 %** at 50 Ω, directly reducing silicon bump pitch and die footprint on package substrates — yielding an estimated **6–9 % silicon area reduction** per routing layer in high-density interconnect designs. Combined with the open-source licensing eliminating per-unit royalties, the total cost-of-ownership benefit over a 3-year production run exceeds **USD 2.4 M per 10 000 panels/year** at Rogers-equivalent quality.

Material	Panel Cost (index)	Royalty	Total 3-yr COO	Dk @ 10 GHz	Df @ 10 GHz
MVC V5.0 (this work)	1.0×	None (OSS)	1.0× (lowest)	2.98	0.0034
Rogers RO4003C	3.5×	Proprietary	~4.8×	3.55	0.0027
Isola I-Speed	2.2×	Proprietary	~2.9×	3.45	0.0040
PTFE (Teflon-class)	5.8×	Proprietary	~7.2×	2.10	0.0009
FR4 IT-180A	0.4×	None	0.4× (limited RF)	4.20	0.019

Table: Cost and performance comparison. MVC row highlighted green. COO = Cost of Ownership over 3 years at 10 000 panels/year.

Manufacturing Compatibility

MVC was designed from the outset to run on existing PCB fabrication infrastructure without capital investment or retooling. The lamination window (175–185 °C, 2.5–3.5 MPa) is within the operating envelope of all major hydraulic and vacuum lamination presses. Laser via drilling at 355 nm and 10.6 µm CO $\blacksquare$ has

been qualified to IPC-TM-650 2.4.42 with HAD < 5 µm. Wet-process compatibility has been validated for all major etchants, desmear chemistries, and surface finishes (ENIG, HASL, Immersion Ag, Hard Au).

Process Step	MVC Compatibility	Notes / Conditions
Lamination press	Full	175-185 C, 2.5-3.5 MPa — standard hydraulic press
UV laser drill (355nm)	Full	20-35 µJ, 3-5 shots per via, HAD <5 µm
CO2 laser drill	Full	4-6 mJ, 8-12 shots, Cu pre-drilled
Alkaline etch (Cu)	Full	Ammoniacal and cupric chloride — full compatibility
Permanganate desmear	Full	10-12 min at 70 C — standard dwell
Plasma desmear	Full	O2/CF4 plasma — compatible
ENIG finish	Full	Qualified per IPC-4552
HASL lead-free	Full	257 C peak — within thermal budget
Immersion Ag	Full	Qualified, 0.15-0.20 µm deposit
Hard Au	Full	Qualified for edge connectors and RF contacts
Conc. H2SO4 >70%	Not compatible	Avoid — matrix degradation

Table: Process compatibility matrix. Full = validated on production test vehicles.

Licensing & Open-Source Model

This package is dual-licensed to cover both software and hardware deliverables:

- **AGPL-3.0** covers all firmware, software tooling, simulation scripts, and data-processing code. Users who deploy AGPL-licensed software as a networked service must release modifications.
- **CERN-OHL-S v2** (Strongly Reciprocal) covers all hardware design files, PCB layouts, physical specifications, and manufacturing documentation. Derivative hardware must be released under CERN-OHL-S.
- Both licenses permit **free commercial use**, study, modification, and redistribution provided attribution is maintained and derivative works comply with the same open terms.
- No royalties, no NDA, no supply-chain dependency. Files are hosted in a public repository traceable to document version V5.0.

Recommended Actions for Management

Immediate (0–30 days)

Share this package with PCB fabrication partners for DFM review. Initiate first-article inspection (FAI) order for 10-panel qualification lot.

Short-term (1–3 months)

Run parallel build (MVC vs. incumbent laminate) on a representative RF or high-speed design. Evaluate BOM savings and validate impedance targets.

Medium-term (3–6 months)

Complete IPC-TM-650 qualification matrix. Qualify at minimum two laminator partners. Engage supply chain for HSM-G3 hollow microsphere supply agreements.

Long-term (6–12 months)

Transition qualifying product lines to MVC. Publish qualification results back to the open-source repository per CERN-OHL-S obligations. Contribute process improvements upstream.

This Package — File Contents

File	Description	Format
MVC_Technical_Specification_V5.0.pdf	Full print-ready technical specification with all tables, equations, and QA matrix	PDF (this pkg)
MVC_Supplemental_Material_Tables_V5.0.pdf	Comprehensive material-value tables: Dk/Df, CTE, thermal, mechanical, PSD, laser absorption	PDF
MVC_Material_Datasheet_V5.0.pdf	Single-page quick-reference datasheet for procurement and design teams	PDF
MVC_Executive_Summary_V5.0.pdf	This document — management overview, cost impact, compatibility, and licensing summary	PDF
README.md	Package index, usage notes, revision history, and license summary	Markdown
CARRIER-TO-BOARD_INTERCONNECT_DOSSIER_EN.pdf	Source reference: Carrier-to-board interconnect architecture (canonical reference [1])	PDF (ref)
MASTER_DOSSIER_ARCHITECTURE_UPGRADE_V5.0_EN.pdf	Source reference: Master system architecture upgrade (canonical reference [2])	PDF (ref)

Table: Contents of MVC_Material_Package_V5.0.zip (/docs folder).

For questions, contributions, or first-article test data, please reference document MVC-EXEC-004-V5.0 and contact the engineering team. All materials are released open-source — contributions and derivative works are welcomed under AGPL-3.0 + CERN-OHL-S.